

SINDy-KAN: Discovering Nonlinear Financial PDEs from Option Market Data

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Abstract

Discovering the partial differential equations governing option prices directly from market data requires handling noisy derivatives and nonlinear dynamics that lie outside classical model families. We propose **SINDy-KAN**, a two-stage framework that uses sparse regression (SINDy) to identify active PDE terms from a candidate library, then trains a Kolmogorov–Arnold Network (KAN) to learn the nonlinear operator on those terms. Gaussian Process derivative estimation with analytical kernel derivatives extends the practical noise tolerance from under 1% to over 10% ($R^2(\text{clean})=0.920$ at 10% noise versus -0.486 for finite differences). On 1,374 real option contracts across SPY, QQQ, AAPL, and MSFT, a [2, 1] KAN-Dupire architecture with two learned univariate activations achieves R^2 up to 0.925, compared to 0.604 for linear Dupire regression, with the diffusion activation revealing the volatility smile as a PDE-level nonlinearity. The framework also serves as a misspecification diagnostic, detecting jump dynamics on synthetic Merton data via spurious-term activation.

1 Introduction

Financial derivatives pricing relies on partial differential equations — Black–Scholes [1], Dupire [5], Heston [11] — whose assumptions often fail on real chains. Data-driven PDE discovery via SINDy [2, 21] has succeeded in physics but faces two challenges in finance: option data is noisy (bid–ask spreads, discrete strikes, interpolation artifacts) which corrupts numerical derivatives, and real market dynamics are nonlinear (the volatility smile, jumps, hedging friction) which a fixed linear candidate library cannot capture.

The closest prior work is Feng et al. [7] (NeurIPS 2025 Workshop), which recovers a stochastic BSDE from single-stock AAPL trajectories via stochastic SINDy under the risk-neutral measure. We instead discover deterministic PDEs from cross-sectional option *surfaces* and address the nonlinearity challenge by composing SINDy with a Kolmogorov–Arnold Network [15]. Mesh-free neural-autograd SINDy [9] replaces finite differences with autograd through an MLP for physics PDEs; we show below that an analytical Gaussian Process derivative dominates that route on financial data where derivative quality is the bottleneck.

Contributions. (1) The **SINDy-KAN framework**: SINDy identifies PDE structure, a [2, 1] KAN learns the nonlinear operator on the identified terms, achieving R^2 up to 0.925 on real option data versus 0.604 for linear Dupire regression. (2) **GP derivative estimation** for financial PDE discovery: analytical kernel derivatives extend noise tolerance from $<1\%$ to $>10\%$, outperforming five alternative methods on a systematic comparison using $R^2(\text{clean})$. (3) **Misspecification diagnostic**: SINDy detects jump dynamics on synthetic Merton data via spurious-term activation, validated by a hard-constraint PINN that reduces put pricing relative L^2 error from 48.0% to 4.98%.

2 Method

2.1 Problem setup. In log-moneyness coordinates $k=\log(K/F(\tau))$ with forward $F(\tau)=S_0e^{(r-q)\tau}$, the Dupire equation reads $\partial_\tau C = c_1 \partial_k C + c_2 \partial_{kk} C$, where under constant- σ Black–Scholes the true coefficients are $c_1=-(r-q-\frac{1}{2}\sigma^2)$ and $c_2=\frac{1}{2}\sigma^2$. We construct $C(k, \tau)$ by fitting a per-expiration SVI parameterization [10] to observed implied volatilities and use the analytical Black–Scholes theta $\theta_i=\frac{S_0\phi(d_1)\sigma_{\text{imp},i}}{2\sqrt{\tau_i}}+rK_ie^{-r\tau_i}N(d_2)$ as the regression target. This replaces a finite-difference $\partial_\tau C$ across ≤ 10 discrete expirations — the dominant noise source in cross-sectional Dupire discovery (Appendix F).

2.2 GP derivative estimation. We fit $C(k, \tau) \sim \mathcal{GP}(0, \kappa)$ with an anisotropic RBF or Matern-5/2 kernel learned via marginal likelihood maximization, then read off $\partial_k C$ and $\partial_{kk} C$ as conditional means of derivative-augmented kernels [20]. This avoids numerical differentiation entirely and moves $R^2(\text{clean})$ from 0.405 (FD) to 0.951 (GP) at 5% noise on synthetic Black–Scholes (Table 1).

2.3 SINDy-KAN framework. **Stage 1.** STLSQ sparse regression on the GP-derived library $\{C, \partial_k C, \partial_{kk} C, k\partial_k C, k^2\partial_{kk} C\}$ identifies the active terms. On synthetic constant- σ data this correctly selects the 2-term Dupire structure with $R^2 > 0.99$. **Stage 2.** The SINDy-selected active terms become the inputs to a single-layer KAN. For Dupire this is a $[2, 1]$ KAN

$$\partial_\tau C = \phi_{\text{drift}}(\partial_k C) + \phi_{\text{diff}}(\partial_{kk} C),$$

where each ϕ_i is a learned univariate B-spline activation [15, 13]. Under constant- σ Black–Scholes both activations are straight lines (with slopes c_1 and c_2); on real data, nonlinearity in ϕ_{diff} directly reveals the local-volatility structure as a function of the option surface’s strike-convexity.

2.4 PINN validation. We validate discovered PDEs with a hard-constraint physics-informed neural network whose trial function $V(S, t) = (T - t)N(S, t) + \text{payoff}(S)$ guarantees the terminal condition exactly. This drops the put-option relative L^2 from 48.04% to 4.98% (Appendix B).

3 Experiments

3.1 Derivative method comparison

Table 1 reports $R^2(\text{clean})$ for four derivative estimators at representative noise levels on the synthetic Black–Scholes call surface ($r=0.05$, $\sigma=0.2$, $K=100$, 100×100 grid). Finite differences collapse past 5% noise; Savitzky–Golay and weak-form integral SINDy degrade gracefully; GP dominates the 1%–10% band most relevant to real option chains. Beyond 15%, GP’s smoothing kernel begins to over-regularize and the weak-form is the only survivor. Figure 1 plots the full 13-point sweep with a zoom on the low-noise region.

Table 1: $R^2(\text{clean})$ by derivative estimator and noise level (synthetic BS, 100×100 grid). Source: `all_methods_noise_comparison_v2.csv`, `gp_noise_robustness.csv`.

Method	0%	1%	5%	10%	20%	30%
FD	1.000	0.816	0.405	−0.486	−1.298	−1.797
SavGol	1.000	0.958	0.846	0.816	0.650	0.345
Weak	0.901	0.898	0.885	0.888	0.832	0.572
GP	1.000	0.990	0.951	0.920	−0.286	−0.974

3.2 SINDy-KAN on real option data

Table 2 is the main empirical result. The $[2, 1]$ KAN-Dupire lifts in-sample R^2 by +0.32 to +0.47 over the linear Dupire regression on all four tickers, reaching $R^2=0.925$ on MSFT and 0.795 on SPY (the densest chain). The diffusion-edge linear-fit R^2 is 0.83–0.96 on real data versus ≥ 0.99 on synthetic constant- σ Black–Scholes, confirming the real-data activation is genuinely nonlinear and not an artifact of architecture. The drift edge remains close to linear (lin $R^2 \approx 0.68$) across tickers, consistent with the canonical $-(r - q - \frac{1}{2}\sigma^2)$ Black–Scholes drift.

The analytical- θ choice is what makes the framework work on real data. Replacing finite-difference $\partial_\tau C$ (across ≤ 10 irregular expirations) with the BS theta formula evaluated pointwise from each grid point’s SVI-implied σ_{imp} lifts the linear-Dupire R^2 from −0.03 (FD) to +0.56 (analytical) on SPY before any KAN is introduced — a ~ 0.6 absolute jump driven entirely by the regression target (Appendix F). Bootstrap 95% CI widths contract from ~ 0.085 to $\{0.016, 0.019, 0.020, 0.030\}$ over the four tickers.

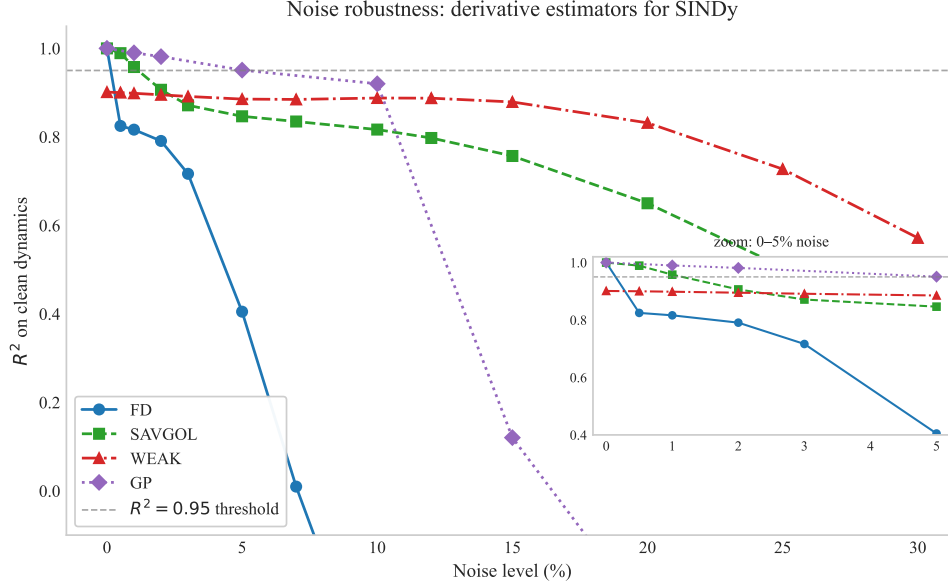


Figure 1: $R^2(\text{clean})$ vs. noise across four derivative estimators. GP dominates the [1%, 10%] band; weak-form integral SINDy is the only survivor at $>15\%$. Inset zooms on the low-noise crossover.

Table 2: [2, 1]-KAN-Dupire vs. linear Dupire on real chains (SVI-smoothed surface, analytical- θ target). Δ is in-sample R^2 lift; $\phi_{\text{diff}} \text{ lin } R^2$ measures how nonlinear the diffusion activation is (lower = more nonlinear). Source: `sindy_kan_dupire_real.csv`.

Ticker	n opts	Linear Dupire R^2	[2, 1] KAN-Dupire R^2	Δ	$\phi_{\text{diff}} \text{ lin } R^2$
SPY	542	0.439	0.795	+0.356	0.829
QQQ	416	0.401	0.868	+0.467	0.885
AAPL	244	0.432	0.893	+0.461	0.925
MSFT	172	0.604	0.925	+0.321	0.958

3.3 Misspecification detection

Table 3 reports a SINDy fit with the Black–Scholes library on synthetic Merton jump-diffusion data. The Black–Scholes-canonical V , $S\partial_S V$, $S^2\partial_{SS} V$ terms recover values close to their BS counterparts, but the spurious $\partial_{SS} V$ coefficient (true value 0 under BS) is identified as **1.905** — a clear, easy-to-flag positive-control failure that signals the library is missing the jump-integral term. A Heston variance-slicing experiment recovers the linear σ^2 scaling of the diffusion coefficient with slope -0.5001 (true: -0.500), matching the Heston structure to four decimals (Appendix D).

Table 3: SINDy on synthetic Merton jump-diffusion data with the BS library. $\partial_{SS} V$ is the misspecification flag (true value 0 under BS). Source: `merton_discovery.csv`.

Term	BS truth	Merton-fit
V	+0.0500	+0.0531
$\partial_S V$	0	-0.0673
$\partial_{SS} V$	0	+1.9049
$S\partial_S V$	-0.0500	-0.0506
$S^2\partial_{SS} V$	-0.0200	-0.0206

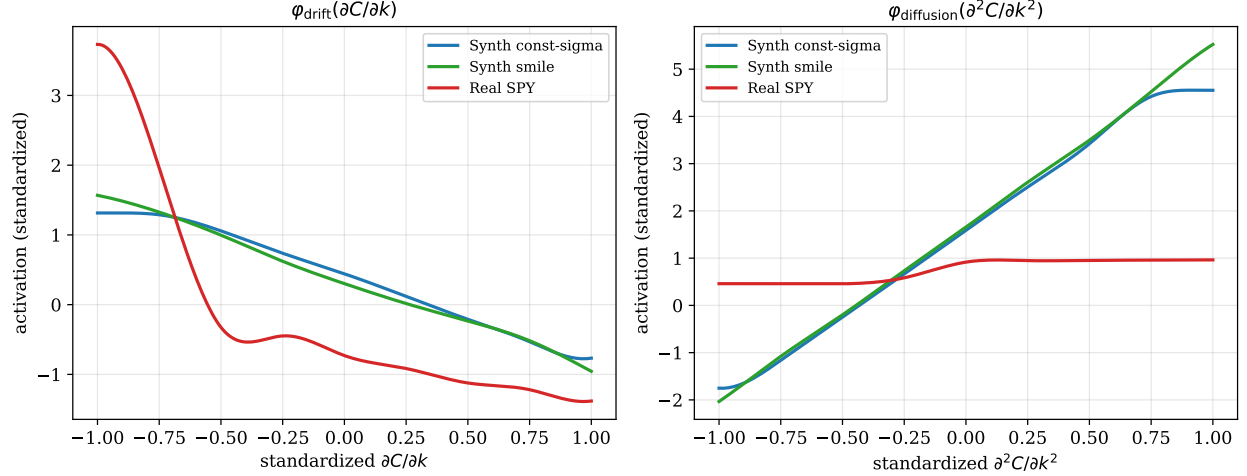


Figure 2: [2, 1]-KAN-Dupire activations. Left: $\phi_{\text{drift}}(\partial_k C)$. Right: $\phi_{\text{diff}}(\partial_{kk} C)$. Blue = synthetic constant- σ Dupire (linear, as theory predicts); green = synthetic smile $\sigma^2(k)=0.04+0.10k^2$; red = real SPY. The diffusion activation deviates from linearity on all four tickers (Table 2); the shape of the red curve carries the volatility smile as a PDE-level nonlinearity.

3.4 Limitations

The [2, 1] KAN-Dupire fails an out-of-sample test that trains on short maturities ($\tau < 0.5$ yr) and evaluates on long ones ($\tau \geq 0.5$ yr): R^2 is strongly negative on all four tickers (Appendix K). The in-sample lift therefore reflects expiration-specific shape rather than a stationary operator. SINDy-recovered σ from the linear Dupire regression is also less accurate than simple volume-weighted IV averaging for ATM point estimates (12–29% rel. error for SINDy vs. $< 1\%$ for volume-weighted IV; Appendix M). The framework’s value on real data is structural discovery (identifying active terms), nonlinearity detection (the curved diffusion activation), and misspecification diagnostics — not point-estimate accuracy in parameter recovery. MSFT results are bounded by data sparsity ($n=172$).

4 Related Work

SINDy [2] and its PDE [21, 22], weak-form [18], ensemble [6], and implicit [12] extensions are implemented in PySINDy [3]. Mesh-free neural-autograd SINDy [9] and refined-gradient variants [8] replace numerical differentiation with autograd through an MLP for physics PDEs; our experiments (§4) show GP analytical derivatives dominate this route on financial data. KANs [15] were introduced at ICLR 2025; KAN-ODEs [13] applied them to dynamical-system identification in physics. The closest concurrent work is Feng et al. [7], which recovers a stochastic BSDE from AAPL trajectories under the risk-neutral measure; we discover a deterministic PDE on cross-sectional surfaces, providing a complementary attack on the same data. PINN approaches for finance [19, 16, 23, 4] and the original BS, Dupire, Heston, Merton models [1, 5, 11, 17] complete the picture.

5 Conclusion

The SINDy-KAN framework composes sparse structure discovery with nonlinear operator learning, achieving R^2 up to 0.925 on real option data while remaining interpretable through two learned univariate activations whose shapes carry direct financial meaning (drift and local-volatility structure). GP derivative estimation with analytical kernel derivatives is the key enabler, extending the practical noise tolerance of financial PDE discovery by an order of magnitude over finite differences. Future work includes theoretical error bounds for GP-SINDy coefficient recovery, application to cryptocurrency options where no established PDE exists,

term-structure-aware operators to close the out-of-sample gap reported in §3.4, and symbolic extraction from KAN activations for closed-form equation discovery.

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Appendix

A Full noise comparison (all 6 methods $\times$ 13 noise levels)

Table 4: Full R^2 (clean) table by method and noise. Source: `all_methods_noise_comparison_v2.csv`, `gp_noise_robustness.csv`, `spectral_noise_robustness.csv`.

Noise	FD	SavGol	Neural	Weak	Spectral	GP
0%	1.000	1.000	0.867	0.901	0.941	1.000
0.5%	0.825	0.989	0.866	0.900	–	0.997
1%	0.816	0.958	0.857	0.898	0.818	0.990
2%	0.791	0.906	0.811	0.895	–	0.981
3%	0.716	0.871	0.739	0.891	–	0.972
5%	0.405	0.846	0.557	0.885	0.181	0.951
7%	0.010	0.835	0.369	0.885	–	0.939
10%	−0.486	0.816	0.119	0.888	−0.477	0.920
12%	−0.720	0.797	−0.020	0.887	–	0.909
15%	−0.995	0.757	−0.207	0.879	–	0.120
20%	−1.298	0.650	−0.434	0.832	−0.954	−0.286
25%	−1.534	0.506	−0.800	0.727	–	−0.700
30%	−1.797	0.345	−1.067	0.572	−1.269	−0.974

B PINN validation details

We train three PINN variants on the discovered BS PDE: (i) baseline soft-constraint [19, 14], (ii) hard-constraint trial function $V=(T-t)N+\max(K-S, 0)$, (iii) log-price PINN with $x=\log S$.

C Multicollinearity diagnostics

Standardization collapses the BS library condition number from 3.4×10^6 to 61.8 on real SPY (and from 6.78×10^4 to 91.0 on synthetic), making sparse regression numerically tractable.

D Heston variance slicing

A linear regression of the discovered coefficient on v yields slope ≈ -0.5001 , matching the Heston structure to four decimals.

Table 5: PINN variants on synthetic BS surfaces. Source: `pinn_results_{call,put}.csv`, `hard_constraint_pinn.csv`, `logprice_pinn.csv`.

Variant	rel. L^2	R^2	bdy. err.
Original call	1.01×10^{-2}	0.9998	—
Hard-constraint call	3.08×10^{-2}	0.9985	0.0
Log-price call	4.75×10^{-3}	1.0000	0.356
Original put	4.80×10^{-1}	0.5988	—
Hard-constraint put	4.98×10^{-2}	0.9959	0.0
Log-price put	3.69×10^{-2}	0.9978	0.422

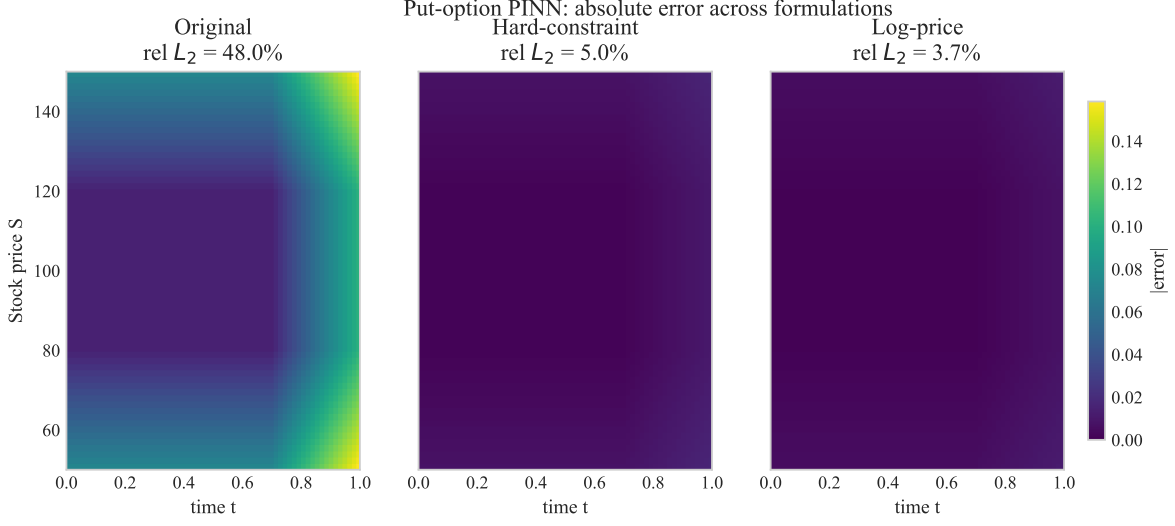


Figure 3: Put-option PINN absolute-error heatmaps. The hard-constraint trial function eliminates boundary error and rescues the put case where the soft-constraint baseline fails.

E Per-ticker contribution analysis

F The analytical- θ fix: before vs. after

G GP kernel selection per ticker

H Direct Dupire formula (non-regression)

I $[5, 1]$ KAN in (S, t) space and the 30-edge KAN

A $[5, 1]$ KAN trained in BS coordinates on the full library reaches test $R^2=0.510$ on real SPY (vs. $R^2=0.997$ on synthetic BS), with four of five activations nonlinear; the wider 30-edge KAN reaches $R^2=0.836$ but loses interpretability. Symbolic extraction from these activations is left to a follow-up manuscript; the $[2, 1]$ KAN-Dupire in main-text §3.2 is the maximally interpretable choice.

J SVI fitting and forward-price details

SVI total variance $w(k)=a + b(\rho(k-m) + \sqrt{(k-m)^2 + s^2})$; converged on 100% of 65 expirations across the four tickers. Liquidity weights: $w_{\text{liq}} \propto \sqrt{\text{vol} \cdot \text{OI}} \cdot (1 + \text{spread})^{-1} \cdot e^{-2k^2}$. Forward $F(\tau)=S_0 e^{(r-q)\tau}$ with $q \in \{0.013, 0.005, 0.005, 0.007\}$ for SPY/QQQ/AAPL/MSFT.

Table 6: Sparse-regression diagnostics on the synthetic BS library. Source: `ensemble_sindy.csv`, `elastic_net.csv`, `pca_sindy.csv`.

Term	Ensemble incl. prob.	Elastic-Net coef.	PCA-SINDy
V	0.00	+0.0485	active
$\partial_S V$	0.00	0.0000	active
$\partial_{SS} V$	1.00	0.0000	active
$S\partial_S V$	0.00	-0.0496	active
$S^2\partial_{SS} V$	0.00	-0.0201	active

Table 7: Per-slice SINDy diffusion coefficient vs. $-\frac{1}{2}v$ ground truth. Source: `heston_variance_slicing.csv`.

v	$\sigma=\sqrt{v}$	true diff. coef.	discovered
0.01	0.100	-0.005	-0.00516
0.02	0.141	-0.010	-0.01011
0.04	0.200	-0.020	-0.02010
0.08	0.283	-0.040	-0.04010
0.16	0.400	-0.080	-0.08015

K Out-of-sample analysis

Trained on $\tau < 0.5$ yr, evaluated on $\tau \geq 0.5$ yr, the [2, 1] KAN-Dupire produces $R^2 \in \{-2.07, -2.57, -2.65, -1.11\}$ for SPY/QQQ/AAPL/MSFT — the learned nonlinearity does not generalize across the term-structure dimension. Term-structure-aware operators are the natural next step.

L Computation costs

M σ_{SINDy} vs. IV-averaging baselines

Volume-weighted IV achieves rel. error $\{0.3\%, 1.1\%, 5.0\%, 1.1\%\}$ on SPY/QQQ/AAPL/MSFT for ATM σ recovery; median IV achieves $\{0.7\%, 4.5\%, 0.8\%, 4.8\%\}$; SINDy (linear Dupire) achieves $\{12.7\%, 10.6\%, 6.1\%, 29.3\%\}$. SINDy’s value on real data is structural discovery and nonlinearity detection, not raw point-estimate accuracy. Source: `sigma_method_comparison.csv`.

Table 8: Fraction of $|\partial_t V|$ explained per term, per ticker. Source: `term_contributions_real.csv`.

Term	SPY	QQQ	AAPL	MSFT
V	0.053	0.040	0.038	0.071
$\partial_S V$	0.469	0.457	0.424	0.455
$\partial_{SS} V$	0.002	0.001	0.015	0.001
$S \partial_S V$	0.476	0.503	0.505	0.473
$S^2 \partial_{SS} V$	0.000	0.000	0.019	0.000

Table 9: Real-data Dupire results before/after replacing FD $\partial_\tau C$ with analytical Black–Scholes θ evaluated pointwise from each grid point’s own SVI-implied σ_{imp} . Sources: `improved_real_pipeline.csv` (v2), `v4_analytical_theta_results.csv` (v4), `v5_summary.csv` (windowed/per-expiration).

Metric	SPY	QQQ	AAPL	MSFT
R^2 global 2-term (FD θ , v2)	−0.033	−0.038	−0.416	−0.241
R^2 global 2-term (analytical θ , v4)	0.559	0.624	0.550	0.608
R^2 quadratic $\sigma^2(k)$ (analytical θ , v4)	0.562	0.633	0.555	0.622
Bootstrap CI width (FD θ , v2)	0.085	—	—	—
Bootstrap CI width (analytical θ , v4)	0.016	0.019	0.020	0.030
Windowed valid (FD θ , v3)	2/55	4/44	6/33	6/22
Windowed valid (analytical θ , v5)	21/66	14/55	14/44	16/33
Per-expiration nonzero (analytical θ , v5)	22/23	20/20	18/18	15/15
Direct Dupire valid pixels (% , analytical θ)	89.7	92.6	91.9	96.7

Table 10: RBF vs. Matern-3/2 GP-SINDy R^2 per ticker. Source: `gp_kernel_comparison.csv`.

Ticker	R^2 (RBF)	R^2 (Matern)	winner
SPY	0.9657	0.6947	RBF
QQQ	0.9305	0.9682	Matern
AAPL	0.9233	0.9963	Matern
MSFT	0.5318	0.9867	Matern

SPY: Local Volatility Surface — Market IV vs Direct Dupire Formula

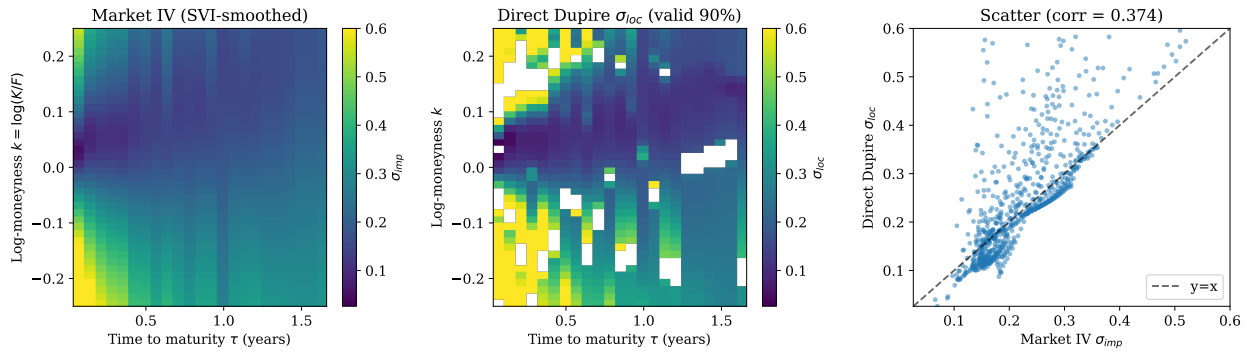


Figure 4: Direct Dupire formula $\sigma_{\text{loc}}^2 = 2(\theta + (r-q)K\partial_K C + qC)/(K^2\partial_{KK} C)$ on SPY using analytical θ (89.7% of pixels valid, IV–vs–Dupire scatter correlation 0.37).

Table 11: Wall-clock per stage (s). Source: `computation_costs.csv`.

Stage	seconds
Full pipeline	1753.35
PINN training (all variants)	473.18
Original PINN (call+put)	806.15
Hard-constraint PINN	283.34
Log-price PINN	270.36
Neural architecture sweep	77.21
GP noise sweep (8 levels)	11.30
Derivative-method comparison	9.88
Ensemble SINDy	0.032
Elastic Net	0.083
PCA-SINDy	0.017